

LEARNING TASK 2

Course ID: Probability & Statistics

Instructor: Dr. ir. D.A.M.P. Blom

Independence

Two events A and B are defined to be *independent* if and only if:

$$P(A|B) = P(A) \quad \text{or} \quad P(B|A) = P(B)$$

Corollary If A and B are independent, then the probability of their joint occurrence satisfies:

$$P(A \cap B) = P(A) \cdot P(B)$$

This property follows directly from probability rule R.4.1.

Independence of Complements If A and B are independent events, then their complement events A' and B' are also independent.

Proof Using the addition rule for probabilities from the previous learning task, we get:

$$\begin{aligned} P(A' \cap B') &= P((A \cup B)') \\ &= 1 - P(A \cup B) \\ &= 1 - P(A) - P(B) + P(A \cap B) \\ &= 1 - P(A) - P(B) + P(A) \cdot P(B) \\ &= (1 - P(A)) \cdot 1 - (1 - P(A)) \cdot P(B) \\ &= (1 - P(A)) \cdot (1 - P(B)) \\ &= P(A') \cdot P(B') \end{aligned}$$

Example: Semiconductor Manufacturing Revisiting the semiconductor manufacturing scenario from the previous learning task, we had $P(B) = 0.024$ and $P(B|A) = 0.1$. Because $P(B|A) \neq P(B)$, events A and B are dependent.

Example: Circuit Reliability Consider a circuit containing three devices: A , B , and C , where failure occurs independently across all components.

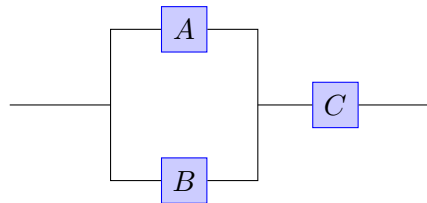


Figure 1: Circuit configuration with three operational devices.

Let the individual operational probabilities be given as $P(A) = 0.95$, $P(B) = 0.95$, and $P(C) = 0.90$. The overall probability that the circuit functions is formulated as:

$$P(\text{circuit functions}) = P((A \cup B) \cap C) = P(A \cup B) \cdot P(C)$$

To compute the parallel subsystem reliability $P(A \cup B)$:

$$\begin{aligned}
 P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\
 &= P(A) + P(B) - P(A) \cdot P(B) \\
 &= 0.95 + 0.95 - (0.95)^2 \\
 &= 0.9975
 \end{aligned}$$

Alternative Method The parallel subsystem probability $P(A \cup B)$ can also be calculated using complementary probabilities:

$$\begin{aligned}
 P(A \cup B) &= 1 - P((A \cup B)') \\
 &= 1 - P(A' \cap B') \\
 &= 1 - P(A') \cdot P(B') \\
 &= 1 - (1 - P(A)) \cdot (1 - P(B)) \\
 &= 1 - (0.05)^2 \\
 &= 0.9975
 \end{aligned}$$

Thus, the overall functional probability of the system is:

$$P(\text{circuit functions}) = 0.9975 \cdot 0.9 = 0.89775$$

Bayes' Theorem

We revisit once again the semiconductor example from the previous learning task. Suppose a semiconductor fails (B). To find the probability that the chip was subjected to high contamination levels during manufacturing (A), we evaluate the inverse conditional probability $P(A|B)$.

When evaluating conditional probabilities in reverse operational order, Bayes' Theorem is applied.

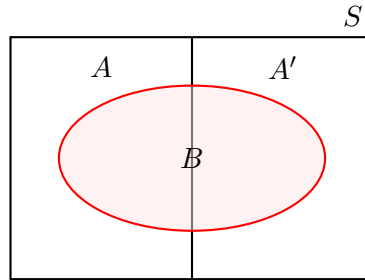


Figure 2: Partitioning of event B across sample space partitions A and A' .

By definition of conditional probability and the Law of Total Probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

Example: Contamination Diagnostics Using the values from semiconductor manufacturing in Example 3:

$$P(\text{high contamination} \mid \text{failure}) = P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{0.10 \cdot 0.20}{0.024} \approx 0.8333$$

Random Variables

Given a random experiment E with associated sample space S (for instance, engaging a target in up to five attempts, $S = \{h, mh, mmh, mmmh, mmmmh\}$ where m = miss and h = hit):

A function X that assigns a real number to each outcome in the sample space S is defined as a **random variable**.

- **Discrete Random Variable:** a random variable with a finite or countably infinite range.
- **Continuous Random Variable:** a random variable whose range consists of an uncountably infinite interval (finite or infinite) of real numbers.

Example 1: Target Engagement In engaging a target up to five attempts, let X represent the attempt number on which the target is eliminated. Here, $X(mmh) = 3$ and $X(h) = 1$. Because the range of X is the finite set $\{1, 2, 3, 4, 5\}$, X is a discrete random variable.

Example 2: Lifespan Analysis Let X denote the operational lifespan of a light bulb. Since the range of $X = \{x \in \mathbb{R} \mid x \geq 0\}$ forms a continuous real interval, X is a continuous random variable.

Example 3: Geometric Trials Consider engaging a target in an infinite sequence of independent attempts with a constant single-trial success probability p . Let X denote the attempt number on which the target is eliminated. Because the range of $X = \{1, 2, 3, \dots\}$ is countably infinite, X is a discrete random variable.